

David Vayntrub

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University of California, Riverside

Computer Science, Expected Graduation: 2027

Experiences:

- **StudySpot – Cofounding Engineer** **May 2025 - Sep 2025**
 - AI-based study tool that uses school documents as context with the Anthropic LLM built using a Next.js and TypeScript frontend and Supabase backend
 - Integrated Canvas LMS API to determine courses and upload documents based on API key, decreasing new user onboarding time by 50%
 - Improved document ingestion speed by 30% by integrating LlamaParse
- **ACM Riverside - Forge – Software Engineer** **Jan 2025 - Mar 2025**
 - Collaborated on a 10-member team to develop a self-moving chess set
 - Utilized Python SpeechRecognizer and PyAudio to detect and parse voice command
 - Achieved a 73% reduction in Word Error Rate (dropping from 15% to 4%) by integrating Python SpeechRecognition optimizations for speech processing.
- **Cutie Hack 2025 – Fullstack Developer** **Nov 2025**
 - Engineered an interactive visual programming environment using Next.js and Blockly, mapping custom block-based logic into executable JavaScript.
 - Designed 15+ custom syntax and control-flow blocks with custom code generators, integrating a sandboxed client-side runtime for real-time script evaluation and instant visual feedback.

Projects:

- **Analysis of Machine Learning Methods with Regression** **Aug 2022 - Mar 2024**
 - Created a classification pipeline across 32,560+ census records to predict adult income brackets using Python's numpy and pandas libraries to visualize and handle data
 - Achieved a prediction accuracy of 78.01% with Logistic Regression after evaluating Decision Trees, KNN, and SVM in scikit-learn using 5-fold cross-validation.
 - Wrote 20-page research paper detailing methodology and evaluating algorithm effectiveness
- **Personal Website** **May 2026 - Present**
 - Architected a multimodal portfolio featuring typed, voice, and visual search with fuzzy ranking and typo tolerance for instant content discovery.
 - Eliminated redundant LLM API calls and cut query latency by implementing a client-side session cache, pre-rendered summaries, and streamed Gemini API responses.
- **Open Source Poker Solver** **Jul 2026 - Present**
 - Engineered a full-stack Heads-Up No-Limit Hold'em trainer integrating a high-performance Rust CFR/neural solver with a Next.js interactive gameplay interface.
 - Achieved 95.2% action-EV precision and 100% policy-lookup coverage by developing compressed model inference, exact card removal logic, and deterministic betting state trees.
 - Built real-time EV feedback systems, adaptive drill modes, and IndexedDB session persistence, maintaining reliability across 114 automated tests.

Skills:

Languages: Python, TypeScript, JavaScript, Rust, C++, Java, SQL, HTML/CSS

Frameworks & Libraries: React, Next.js, Tailwind CSS, scikit-learn, pandas, NumPy, Google Blockly, PyAudio, SpeechRecognition

Databases & Backend: Supabase (PostgreSQL), IndexedDB

Developer Tools & APIs: Git, Figma, Jupyter Notebooks, Gemini API, Anthropic API, LlamaParse, REST APIs